DRUGS · THYROID AND PARATHYROID

Levothyroxine and liothyronine

Levothyroxine (T4) replaces thyroid hormone after total thyroidectomy and in hypothyroidism, and in differentiated thyroid cancer it is also given in doses that keep TSH low, to reduce TSH-driven growth of residual cancer cells. Liothyronine (T3) acts faster and for a shorter time and is used in selected situations. Every thyroid surgeon must be able to start replacement, interpret the TSH and choose the TSH target by cancer risk.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

LEVOTHYROXINE AND LIOTHYRONINE

	SUMMARY
Levothyroxine	Synthetic T4, half-life about 7 days; converted in tissues to active T3
Liothyronine	Synthetic T3, half-life about 1 day; quick onset and offset
ENT uses	Replacement after total or completion thyroidectomy, and after lobectomy if hypothyroid; TSH suppression in differentiated thyroid cancer; bridging before radioiodine (liothyronine, now largely replaced by recombinant TSH)
Starting dose	After total thyroidectomy in a healthy adult: about 1.6 micrograms/kg daily (for example 100 to 125 micrograms); elderly or with heart disease: 25 to 50 micrograms daily, increased every 4 to 6 weeks
Monitoring	TSH (and free T4) 6 to 8 weeks after starting or changing the dose, then every 6 to 12 months
Taking it	On an empty stomach, 30 to 60 minutes before breakfast, separate from calcium, iron and antacids by 4 hours
Pregnancy	Essential; dose usually needs to rise by 25% to 30% from early pregnancy

2 CHAPTER TWO Using them in ENT

Practice.

2.1 TSH targets in differentiated thyroid cancer

Targets follow the risk of persistent or recurrent disease and the response to treatment (ATA and BTA guidance):

- **High risk** or structural persistent disease: TSH below 0.1 mU/l.
- **Intermediate risk** or biochemically incomplete response: TSH about 0.1 to 0.5 mU/l.
- **Low risk** with excellent response: TSH in the low normal range (about 0.5 to 2 mU/l).

Avoid long-term suppression where it is not needed, particularly in older patients, those with atrial fibrillation, heart disease or osteoporosis.

2.2 After lobectomy

About one in five patients becomes hypothyroid after hemithyroidectomy; check TSH at 6 to 8 weeks.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Over-replacement or suppression: **atrial fibrillation**, palpitations, angina, tremor, weight loss, insomnia and, long-term, **osteoporosis** (particularly in post-menopausal women). Under-replacement: fatigue, weight gain, cold intolerance, raised cholesterol.

3.2 Interactions

Absorption reduced by calcium carbonate, iron, aluminium and magnesium antacids, PPIs, sevelamer, bile acid sequestrants, soy and coffee; requirements increased by enzyme inducers (carbamazepine, rifampicin, phenytoin) and oestrogens; it increases the effect of warfarin.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Starting dose after **total thyroidectomy** and when to check the TSH.
- TSH targets by cancer risk.
- The calcium interaction after thyroidectomy (patients often take both).

4.2 Common pitfalls

COMMON PITFALLS

- Taking levothyroxine with calcium supplements.
- Suppressing TSH indefinitely in a low-risk patient with osteoporosis.

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